

Worksheet — Pendulum Tension Simulation

Simulation 09 · use with *Simulation - Pendulum tension.html* · tension, circular motion (Block 5) and energy (Block 10)

Name: _____ Class: _____ Date: _____

How to use this worksheet

Open **Simulation - Pendulum tension.html** in a browser (it works offline). Choose the object, its size, the string length and θ_0 .  **Hold aside** holds the bob still;  **Release** lets it swing. The newton meter at the top shows the tension.

Part 1 before the lab (≈ 15 min) · **Part 2** with a newton-meter pendulum (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use $g = 9.81 \text{ N kg}^{-1}$. L is measured to the **centre** of the bob.

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. A 200 g mass is held still at 30° by a horizontal pull. Is T less than, equal to or more than mg ?			
2. Released from 30° : T at the bottom of the swing compared with mg ?			
3. Triple the string length, same 30° release. The peak tension?			
4. Double the mass. Peak tension? Period?			
5. Ping-pong ball or steel ball (same size): which keeps swinging longer?			

6. Draw the free-body diagram of the 200 g mass held still at 30° , and calculate the tension and the hand's pull.

Free-body diagram and working

Part 2 — During the lab: newton-meter pendulum (200 g)

situation	sim T / N	our reading / N	theory
at rest			mg
held at 20°			$mg / \cos 20^\circ$
held at 40°			$mg / \cos 40^\circ$
bottom of a swing from 40°			$mg(3 - 2 \cos 40^\circ)$

7. Why is the peak tension of a swing so hard to read on a real newton meter? How could you measure it better?

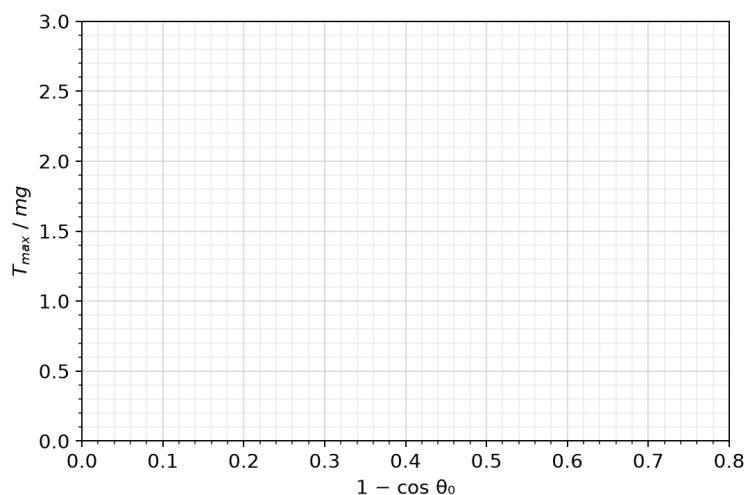
Part 3 — Homework

8. Use energy and the circular-motion equation to show that the tension at the bottom of a swing from θ_0 is $T = mg(3 - 2 \cos \theta_0)$. Why does it not depend on L ?

Derivation

$\theta_0 / ^\circ$	10	20	30	40	50	60	70
$T_{\max} / \text{N}$							
$T_{\max}/mg$							
$1 - \cos \theta_0$							

9. Plot $T_{\max}/mg$ against $(1 - \cos \theta_0)$. Gradient = _____ Intercept = _____



10. Block 10 pendulum: $L = 1.0 \text{ m}$ released from 25° . Calculate the speed at the bottom and the tension there for a 1.0 kg bob. Then calculate the period for small swings.

Working

11. Explain why the balloon pendulum stops after a few swings while the steel ball keeps going.
